AI POWERED SMART ORCHID CARE SYSTEM USING MULTI-MODAL MACHINE LEARNING

Problem

Vanda orchids are widely grown in tropical regions like Sri Lanka, but cultivation remains a manual, expertise-driven process. Growers face four key challenges that lack an integrated digital solution:

1. **Disease:** No accessible, unified system to detect, grade severity, and recommend treatments for local conditions .
2. **Watering & Fertilization:** The exposed aerial roots are easily damaged by reactive watering schedules . Proper care requires precise timing and nutrient balance .
3. **Growth stage & Bloom Prediction:** Flowering timing is difficult to predict. Research on Vanda flowering physiology is limited , forcing growers to rely on guesswork for harvest and sales planning.
4. **Pollination:** Hybrid breeding is largely trial-and-error without compatibility guidance, risking wasted seasons on poor pairings .

Stakeholders

- Commercial growers/nurseries - Need disease control, correct watering/fertilizer, bloom timing, breeding guidance at scale
- Hobbyist growers - Want simple, phone-based care guidance
- Orchid breeders - Need compatibility guidance before hand-pollinating
- Buyers/florists/exporters - Depend on predictable flower availability

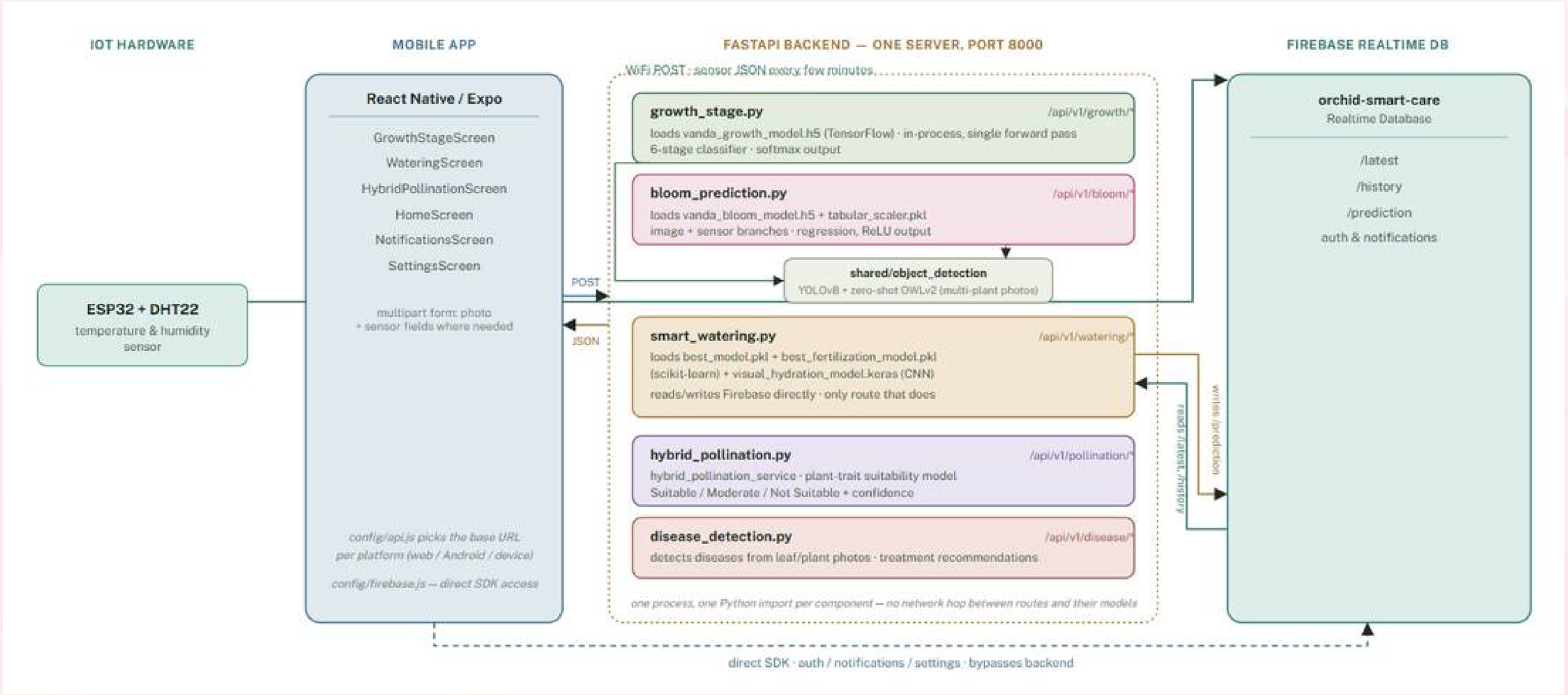
Research Gap and Objectives

Component	Research Gap	Objective
Hybrid Pollination	No data-driven tool predicts parent compatibility. breeders rely on guesswork	Score compatibility from parent traits and give beginners step by step pollination guidance
Disease Detection	Prior work does detection or severity or risk alerts. never all three together for Vanda	Detect disease, grade severity, and recommend treatment in one pipeline
Smart Watering & Fertilization	Existing systems react after moisture drops, none forecast water need and automate fertigation	Predict water/nutrient needs in advance and auto-trigger watering + fertigation
Growth Stage & Bloom Prediction	Closest work needs costly hyperspectral gear, targets Phalaenopsis, no repeat-cycle bloom model	Classify Vanda growth stages and predict bloom timing from a phone photo and basic sensors

Solution Overview

- One mobile app connects to one central server that all four components share.
- **Hybrid Pollination** — upload a photo of two parent plants along with some basic details, and get a compatibility score (Suitable / Moderate / Not Suitable) plus step-by-step breeding guidance.
- **Disease Detection** — upload a plant photo along with environmental readings, and the system detects the disease, tells you how severe it is, and recommends treatment.
- **Smart Watering & Fertilization** — the system predicts when a plant needs water or fertilizer and can trigger watering/feeding automatically, using sensor readings.
- **Growth Stage & Bloom Prediction** — upload a plant photo to identify its current growth stage, and combine it with basic environmental readings to predict how many days until it blooms.

Architecture & integration diagram



Integration & Completion Status

- Hybrid Pollination is fully working end-to-end right now.
- Disease Detection is fully working end-to-end right now.
- Smart Watering & Fertilization is fully working end-to-end right now.
- Growth Stage & Bloom Prediction is fully working end-to-end right now.

Commercialization

- Commercialized as a smart orchid management platform for nurseries and growers.
- Revenue through subscription-based software services.
- Selling IoT monitoring kits and smart devices.
- Licensing the system to nurseries and greenhouse operators.
- Maintenance and support services.
- Future mobile/web app premium features.

Demo roadmap

- Hybrid Pollination
- Disease Detection
- Smart Watering & Fertilization
- Growth Stage & Bloom Prediction



Thank You